

Docker is a platform designed to help developers build, deploy, and manage applications inside lightweight, portable containers. A **Docker container** packages an application with its dependencies, libraries, and configuration files so it can run consistently across different environments.

Key Features:

- **Portability:** Containers run consistently across machines, whether it's a developer's laptop, a test server, or a production cluster.
- **Efficiency:** Containers share the host operating system kernel, making them lightweight compared to virtual machines.
- **Isolation:** Each container runs in its isolated environment, ensuring no interference between applications

Basic Docker Workflow

1. **Build:** Create container images from a **Dockerfile**.
2. **Run:** Start containers from images.
3. **Stop and Remove:** Manage running containers.
4. **Push and Pull:** Share container images through Docker Hub or other registries.

```

[ec2-user@ip-172-31-84-217 ~]$ docker --version
Docker version 25.0.5, build 50c9b7c
[ec2-user@ip-172-31-84-217 ~]$ sudo systemctl enable docker
Created symlink from /etc/systemd/system/multi-user.target.wants/docker.service to /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-84-217 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-84-217 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2024-12-17 08:07:13 UTC; 11s ago
     Docs: https://docs.docker.com
   Process: 3507 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Process: 3506 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
  Main PID: 3510 (dockerd)
    Tasks: 8
   Memory: 29.7M
   CGroup: /system.slice/docker.service
           └─3510 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:6553...

Dec 17 08:07:13 ip-172-31-84-217.ec2.internal systemd[1]: Starting Docker Application Container Engine...
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal dockerd[3510]: time="2024-12-17T08:07:13.212594309Z" level=info msg="Star... up"
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal dockerd[3510]: time="2024-12-17T08:07:13.261549804Z" level=info msg="Load...rt."
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal dockerd[3510]: time="2024-12-17T08:07:13.470664567Z" level=info msg="Load...ne."
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal dockerd[3510]: time="2024-12-17T08:07:13.483955442Z" level=info msg="Dock...0.6
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal dockerd[3510]: time="2024-12-17T08:07:13.484493180Z" level=info msg="Daem...ion"
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal dockerd[3510]: time="2024-12-17T08:07:13.531463321Z" level=info msg="API ...ock"
Dec 17 08:07:13 ip-172-31-84-217.ec2.internal systemd[1]: Started Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full
[ec2-user@ip-172-31-84-217 ~]$ sudo usermod -a -G docker ec2-user
[ec2-user@ip-172-31-84-217 ~]$ sudo chmod 666 /var/run/docker.sock
[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY      TAG          IMAGE ID      CREATED      SIZE
[ec2-user@ip-172-31-84-217 ~]$ docker pull nginx
Using default tag: latest
Error response from daemon: pull access denied for nginx, repository does not exist or may require 'docker login': denied: req
uested access to the resource is denied
[ec2-user@ip-172-31-84-217 ~]$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over
to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is re
quired for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/

Username: █

```

```
[ec2-user@ip-172-31-84-217 ~]$ docker login
```

Authenticating with existing credentials...

WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/engine/reference/commandline/login/#credentials-store>

Login Succeeded

```
[ec2-user@ip-172-31-84-217 ~]$ docker pull nginx
```

Using default tag: latest

latest: Pulling from library/nginx

bc0965b23a04: Pull complete

650ee30bbe5e: Pull complete

8cc1569e58f5: Pull complete

362f35df001b: Pull complete

13e320bf29cd: Pull complete

7b50399908e1: Pull complete

57b64962dd94: Pull complete

Digest: sha256:fb197595ebe76b9c0c14ab68159fd3c08bd067ec62300583543f0ebda353b5be

Status: Downloaded newer image for nginx:latest

docker.io/library/nginx:latest

```
[ec2-user@ip-172-31-84-217 ~]$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nginx	latest	66f8bdd3810c	2 weeks ago	192MB

```
[ec2-user@ip-172-31-84-217 ~]$ docker run -it 66f8bdd3810c /bin/bash
```

root@1049f99f4555:/# docker ps

bash: docker: command not found

root@1049f99f4555:/# docker ps -a

bash: docker: command not found

root@1049f99f4555:/# exit

exit

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
1049f99f4555	66f8bdd3810c	"/docker-entrypoint...."	About a minute ago	Exited (127)	16 seconds ago	wonderful_

```
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name development 66f8bdd3810c
```

b753d13ae9c91f6b2dcc166af3aaa9b42fa8cb0f47e43a075823bf941e9ed05e

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b753d13ae9c9	66f8bdd3810c	"/docker-entrypoint...."	28 seconds ago	Up 27 seconds	80/tcp	development

```
[ec2-user@ip-172-31-84-217 ~]$
```

```
[ec2-user@ip-172-31-84-217 ~]$ docker exec -it ae639f2d714a /bin/bash
root@ae639f2d714a:/# read escape sequence
[ec2-user@ip-172-31-84-217 ~]$ docker kill ae639f2d714a
ae639f2d714a
[ec2-user@ip-172-31-84-217 ~]$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS        NAMES
b753d13ae9c9   66f8bdd3810c        "/docker-entrypoint...." 5 minutes ago  Up 5 minutes  80/tcp       development
[ec2-user@ip-172-31-84-217 ~]$ docker pull iron/images
Using default tag: latest
latest: Pulling from iron/images
[DEPRECATION NOTICE] Docker Image Format v1, and Docker Image manifest version 2, schema 1 support will be removed in an upcoming release. Suggest the author of docker.io/iron/images:latest to upgrade the image to the OCI Format, or Docker Image manifest v2, schema 2. More information at https://docs.docker.com/go/deprecated-image-specs/
a3ed95caeb02: Pull complete
e4019146ac24: Pull complete
d0e0eaf01352: Pull complete
e90dc7db078e: Pull complete
b973f73ba642: Pull complete
2f18d2a5994f: Pull complete
7830a63975a0: Pull complete
9d9857bccbf5: Pull complete
d477d23d9e7f: Pull complete
da8b5094cad7: Pull complete
5e1a73373885: Pull complete
ae99786b8982: Pull complete
6ddcf896ed47: Pull complete
1133d0d97d20: Pull complete
21e4c1c35292: Pull complete
d03b4a51bcf2: Pull complete
687e46735c94: Pull complete
7d1cc49c14ed: Pull complete
56380f39aa52: Pull complete
0a5c126cd125: Pull complete
16fa950142f5: Pull complete
4e54d51ef2df: Pull complete
c3cbb1d7d564: Pull complete
e20b1b3de4b3: Pull complete
1e99653d5098: Pull complete
52fb51f584e8: Pull complete
Digest: sha256:593d0299fd86e891118cb867bc9c86884cf936ea0a4c01dd2ac1448aaa18e7b9
Status: Downloaded newer image for iron/images:latest
docker.io/iron/images:latest
[ec2-user@ip-172-31-84-217 ~]$
```

```
ec2-user@ip-172-31-84-217:~  
[ec2-user@ip-172-31-84-217 ~]$ docker images  
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE  
nginx         latest    66f8bdd3810c   2 weeks ago    192MB  
iron/images    latest    2c203b4f915c   10 years ago    677MB  
[ec2-user@ip-172-31-84-217 ~]$ docker ps  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES  
fa4dba364b2e   2c203b4f915c   "/bin/bash"             5 minutes ago Up 5 minutes   80/tcp         affectionate_hertz  
b753d13ae9c9   66f8bdd3810c   "/docker-entrypoint...." 20 minutes ago Up 20 minutes   80/tcp         development  
[ec2-user@ip-172-31-84-217 ~]$  
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name prod fa4dba364b2e  
Unable to find image 'fa4dba364b2e:latest' locally  
docker: Error response from daemon: pull access denied for fa4dba364b2e, repository does not exist or may require 'docker login'  
n': denied: requested access to the resource is denied.  
See 'docker run --help'.  
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name prod 2c203b4f915c  
docker: Error response from daemon: no command specified.  
See 'docker run --help'.  
[ec2-user@ip-172-31-84-217 ~]$ clear  
[ec2-user@ip-172-31-84-217 ~]$ docker images  
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE  
nginx         latest    66f8bdd3810c   2 weeks ago    192MB  
iron/images    latest    2c203b4f915c   10 years ago    677MB  
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name prod 2c203b4f915c /bin/bash  
0e580077e17583187f25588f39e44275b85f591633fe30fbbb58817b91533ec  
[ec2-user@ip-172-31-84-217 ~]$ docker ps  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES  
0e580077e175   2c203b4f915c   "/bin/bash"             11 seconds ago Up 11 seconds   80/tcp         prod  
fa4dba364b2e   2c203b4f915c   "/bin/bash"             11 minutes ago Up 11 minutes   80/tcp         affectionate_hertz  
b753d13ae9c9   66f8bdd3810c   "/docker-entrypoint...." 26 minutes ago Up 26 minutes   80/tcp         development  
[ec2-user@ip-172-31-84-217 ~]$ docker kill $(docker ps -a -q)  
0e580077e175  
fa4dba364b2e  
b753d13ae9c9  
Error response from daemon: Cannot kill container: b025b0511f8c: container b025b0511f8c7b4ad9002be9d9195f720c9672b584fa6f76de3  
be05431f51e14 is not running  
Error response from daemon: Cannot kill container: ae639f2d714a: container ae639f2d714aa8c6784adf7d749c979c96d63102c1abab9f52f  
cc5933e6b08f6 is not running  
Error response from daemon: Cannot kill container: 1049f99f4555: container 1049f99f455573960331bf3b47874130b896e46ffa6b957eda5  
951dc71e9061c is not running  
[ec2-user@ip-172-31-84-217 ~]$ docker ps  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
```

1.Update Linux Server

`sudo yum update -y`

2. Install Docker

`sudo yum install docker -y`

3.Start Docker Service

`sudo systemctl start docker`

4.Set Permissions to 666 on Docker Socket

`sudo chmod 666 /var/run/docker.sock`

5. Add User to Docker Group

`sudo usermod -aG docker ec2-user`

6.Docker login

username:

Password:

7.Pull an Image from Docker

`docker pull nginx`

8. View Running Containers

`docker ps`

9.View All Containers (Including Stopped)

`docker ps -a`

10.View Docker Images

`docker images`

11. Create a Container from NGINX Image

`docker run -it imageid /bin/bash`(create and go inside the container)

`docker run -itd imageid` (detached)

12. Start a Stopped Container

`docker start <container_id_or_name>`

13. Stop a Running Container

`docker stop <container_id_or_name>`

14. Execute a Command in a Running Container

`docker exec -it <container_id_or_name> /bin/bash`

15. Kill a Container

`docker kill <container_id_or_name>`

16. Delete a Container

`docker rm <container_id_or_name>`

17. Delete an Image

`docker rmi <image_id_or_name>`

18. `docker run -itd --rm --name app3 -p 8084:80 httpd`(once stop the container it will automatically deleted the container)

Docker image:

A **Docker image** is a lightweight, standalone, and executable software package that includes everything needed to run an application. This includes:

- The application code.
- Runtime environment (like Java, Python, or Node.js).
- Required libraries, dependencies, and binaries.
- Configuration files or environment settings

Docker images serve as the blueprint or template for containers. When you run an image, it creates a container—a runtime instance of the image.

How Docker Images Work

- **Build Process:** Images are created using a **Dockerfile**, which contains instructions to assemble the image.
- **Layering:** Images are made up of layers, which are stacked on top of each other. Each layer represents a change (e.g., adding a file or installing a dependency).
- **Storage:** Images are stored in local systems or remote registries like **Docker Hub**, so they can be shared and pulled as needed.

When we use ubuntu os as a image ,then we can able to execute ubuntu command in amazon linux os server.

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-84-217 ~]$ clear
[ec2-user@ip-172-31-84-217 ~]$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
de44b265507a: Pull complete
Digest: sha256:80dd3c3b9c6cecb9f1667e9290b3bc61b78c2678c02cbdae5f0fea92cc6734ab
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   66f8bdd3810c   2 weeks ago    192MB
ubuntu        latest   b1d9df8ab815   3 weeks ago    78.1MB
iron/images    latest   2c203b4f915c   10 years ago   677MB
[ec2-user@ip-172-31-84-217 ~]$ docker run --name app 66f8bdd3810c /bin/bash
255047e5047f01b679e91f00ea5c6f887b1fcade21ff3c6d3ed9b13eed048435
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name app b1d9df8ab815 /bin/bash
docker: Error response from daemon: Conflict. The container name "/app" is already in use by container "255047e5047f01b679e91f00ea5c6f887b1fcade21ff3c6d3ed9b13eed048435". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name web b1d9df8ab815 /bin/bash
f80f5382eb416b438559271dffb2981e75bcbdce4c0a764fcb3ee9b9b9caeed7
[ec2-user@ip-172-31-84-217 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED        STATUS      PORTS     NAMES
f80f5382eb41   b1d9df8ab815   "/bin/bash"   7 seconds ago   Up 7 seconds      80/tcp    web
255047e5047f   66f8bdd3810c   "/docker-entrypoint..."   About a minute ago   Up About a minute      80/tcp    app
[ec2-user@ip-172-31-84-217 ~]$ docker attach f80f5382eb41
root@f80f5382eb41:/# apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package git
root@f80f5382eb41:/# apt update -y
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [707 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [982 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [15.3 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [702 kB]
```

Docker Hub

Docker Hub is a cloud-based registry provided by Docker that allows developers to:

1. **Store** Docker images.
2. **Share** Docker images with teams or the public.
3. **Distribute** pre-built container images for quick deployment.

It acts as the central repository for Docker images, similar to GitHub for code. Docker Hub contains both **official images** maintained by Docker (e.g., `nginx`, `mysql`) and user-contributed images.

Key Features of Docker Hub

1. **Image Hosting:** Store Docker images in public or private repositories.
2. **Official Images:** Verified images for popular software like `nginx` and `mysql`.
3. **Public & Private Repositories:** Share publicly or securely store private images.
4. **Versioning with Tags:** Manage and pull specific versions of images.
5. **Searchable Index:** Find pre-built images quickly.
6. **Integration:** Automate builds with CI/CD pipelines.
7. **Collaboration:** Share images with team members or the public.

General

Tags

Builds

Collaborators

Webhooks

Settings

kechiya/green-image 

Created less than a minute ago

Add a description   INCOMPLETEAdd a category   INCOMPLETE**Docker commands**[Public view](#)

To push a new tag to this repository:

`docker push kechiya/green-image:tagname`**Tags** INCOMPLETE*Pushed images appear here.***Automated builds**

Manually pushing images to Docker Hub? Connect your account to GitHub or Bitbucket to automatically build and tag new images whenever your code is updated, so you can focus your time on creating.

Available with Pro, Team and Business subscriptions. [Read more about automated builds](#) .

[Upgrade](#)

Pushing my images to Docker Hub

```
[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
nginx            latest      66f8bdd3810c  2 weeks ago   192MB
ubuntu           latest      b1d9df8ab815  3 weeks ago   78.1MB
iron/images      latest      2c203b4f915c  10 years ago  677MB
[ec2-user@ip-172-31-84-217 ~]$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
f80f5382eb41   b1d9df8ab815  "/bin/bash"            56 minutes ago  Up 2 minutes  80/tcp         web
255047e5047f   66f8bdd3810c  "/docker-entrypoint..." 57 minutes ago  Up 57 minutes  80/tcp         app
[ec2-user@ip-172-31-84-217 ~]$ docker commit -m "this is new image" -a "shriya" f80f5382eb41 kechiya
sha256:ea5dfb898807b7f666e92e700830bbe8999422c0ebeb635842b57a1dc93ad31b7
[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
kechiya          latest      ea5dfb898807  17 seconds ago 122MB
nginx            latest      66f8bdd3810c  2 weeks ago   192MB
ubuntu           latest      b1d9df8ab815  3 weeks ago   78.1MB
iron/images      latest      2c203b4f915c  10 years ago  677MB
[ec2-user@ip-172-31-84-217 ~]$ docker tag ea5dfb898807 kechiya/green-image:v1
[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
kechiya/green-image  v1          ea5dfb898807  8 minutes ago  122MB
kechiya            latest      ea5dfb898807  8 minutes ago  122MB
nginx              latest      66f8bdd3810c  2 weeks ago   192MB
ubuntu             latest      b1d9df8ab815  3 weeks ago   78.1MB
iron/images        latest      2c203b4f915c  10 years ago  677MB
[ec2-user@ip-172-31-84-217 ~]$ docker login
Authenticating with existing credentials...
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[ec2-user@ip-172-31-84-217 ~]$ docker push kechiya/green-image:v1
The push refers to repository [docker.io/kechiya/green-image]
06a8e0dc6d62: Pushed
687d50f2f6a6: Mounted from library/ubuntu
v1: digest: sha256:e70e5cbd5e32014396edfaad3309ae9d93c5a8a7da8a27805f688d620db64649 size: 741
[ec2-user@ip-172-31-84-217 ~]$
```

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Sort by

Newest ▾



Filter tags

Delete

TAG

■ [v1](#)

Last pushed a minute ago by [kechiya](#)


Digest

OS/ARCH

Last pull

Compressed size ⓘ

[e70e5cbd5e32](#)

linux/amd64

a few seconds ago

58.34 MB

```
docker pull kechiya/green-image:v1
```

Copy

Docker Port Mapping

Map a **host port** to a **container port** to make containerized applications accessible from outside. This allows multiple applications to run on the same server but on different ports.

Isolation: Each application runs in its own container, ensuring independence (no conflicts between applications).

Portability: Applications can be easily moved between environments (local, staging, production).

Efficiency: Containers are lightweight and share the host OS kernel, making them faster and more resource-efficient than virtual machines

Simplicity: Only one IP address to manage, even when running multiple services.

Resource Efficiency: Avoid the overhead of multiple IP addresses and networking interfaces.

Ease of Development: Ideal for local development environments where multiple services need to interact with each other.

Microservices Architecture: Different services (e.g., frontend, backend, database) running on the same host.

API Gateway: Multiple APIs (versioning or services) exposed on different ports.

Web Hosting: Hosting different websites or services on the same server (e.g., <http://localhost:8080> for Site 1, <http://localhost:9090> for Site 2).

Pulled nginx image and created container with host port 8083

```
[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 66f8bdd3810c 2 weeks ago 192MB
[ec2-user@ip-172-31-84-217 ~]$ docker run -it --name webapp1 -p 8083:80 nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2024/12/17 15:05:53 [notice] 1#1: using the "epoll" event method
2024/12/17 15:05:53 [notice] 1#1: nginx/1.27.3
2024/12/17 15:05:53 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2024/12/17 15:05:53 [notice] 1#1: OS: Linux 5.10.228-219.884.amzn2.x86_64
2024/12/17 15:05:53 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2024/12/17 15:05:53 [notice] 1#1: start worker processes
2024/12/17 15:05:53 [notice] 1#1: start worker process 29
2024/12/17 15:05:56 [notice] 1#1: signal 2 (SIGINT) received, exiting
2024/12/17 15:05:56 [notice] 29#29: signal 2 (SIGINT) received, exiting
2024/12/17 15:05:56 [notice] 29#29: exiting
2024/12/17 15:05:56 [notice] 29#29: exit
2024/12/17 15:05:56 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2024/12/17 15:05:56 [notice] 1#1: worker process 29 exited with code 0
2024/12/17 15:05:56 [notice] 1#1: exit
[ec2-user@ip-172-31-84-217 ~]$ docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
95714fb593fb nginx "/docker-entrypoint...." 17 seconds ago Exited (0) 13 seconds ago webapp1
[ec2-user@ip-172-31-84-217 ~]$ docker start 95714fb593fb
95714fb593fb
[ec2-user@ip-172-31-84-217 ~]$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
95714fb593fb nginx "/docker-entrypoint...." 46 seconds ago Up 6 seconds 0.0.0.0:8083->80/tcp, :::8083->80/tcp webapp1
[ec2-user@ip-172-31-84-217 ~]$
```

Inbound rules [Info](#)

Add rule

×

[Preview changes](#)

Save rules

Nginx running on http://34.201.59.201:8083/

← → ↻ 🏠 ⚠ Not secure 34.201.59.201:8083

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```

[ec2-user@ip-172-31-84-217 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
tomcat        latest   f62f518e5c5c   8 days ago    467MB
nginx         latest   66f8bdd3810c   2 weeks ago    192MB
[ec2-user@ip-172-31-84-217 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                                                                 NAME
fc86397b3e59   tomcat    "catalina.sh run"       11 seconds ago Up 11 seconds  0.0.0.0:8081->8080/tcp, :::8081->8080/tcp  appserver
[ec2-user@ip-172-31-84-217 ~]$

```

Inbound rules Info						
Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-034ae147851c7fe0c	SSH ▼	TCP	22	Cu... ▼	0.0.0.0 0.0.0.0/0	Delete
sgr-0bc82298a000ef24f	Custom TCP ▼	TCP	8080	Cu... ▼	0.0.0.0 0.0.0.0/0	Delete
sgr-082a24c36e8457302	Custom TCP ▼	TCP	8081	Cu... ▼	0.0.0.0 0.0.0.0/0	Delete
sgr-05f5c1322815697bb	Custom TCP ▼	TCP	8083	Cu... ▼	0.0.0.0 0.0.0.0/0	Delete

Tomcat is running on http://34.201.59.201:8081/

← → ↻ 🏠 ⚠ Not secure 34.201.59.201:8081 ☆ 📄 k ⋮

HTTP Status 404 – Not Found

Type Status Report

Description The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.

Apache Tomcat/11.0.2

```
[ec2-user@ip-172-31-84-217 ~]$ docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
Digest: sha256:f4c5139eda466e45814122d9bd8b886d8ef6877296126c09b76dbad72b03c336
Status: Image is up to date for httpd:latest
docker.io/library/httpd:latest
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --rm --name app3 -p 8084:80 httpd
d39625c538dfc41acb265d61acfb51b8c4dbd017eb7a0076678295c23c42833b
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
d39625c538df app3	httpd	"httpd-foreground"	5 seconds ago	Up 5 seconds	0.0.0.0:8084->80/tcp, :::8084->80/tcp
70a056e3d179 app2	httpd	"httpd-foreground"	12 minutes ago	Up 12 minutes	80/tcp, 0.0.0.0:9090->90/tcp, :::9090->90/tcp
fc86397b3e59 appserver	tomcat	"catalina.sh run"	23 minutes ago	Up 23 minutes	0.0.0.0:8081->8080/tcp, :::8081->8080/tcp

```
[ec2-user@ip-172-31-84-217 ~]$
```

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-034ae147851c7fe0c	SSH	TCP	22	Cu... Info	0.0.0.0/0	Delete
sgr-082a24c36e8457302	Custom TCP	TCP	8081	Cu... Info	0.0.0.0/0	Delete
sgr-05f5c1322815697bb	Custom TCP	TCP	8083	Cu... Info	0.0.0.0/0	Delete
sgr-06703e205f9ed1a5e	Custom TCP	TCP	8084	Cu... Info	0.0.0.0/0	Delete

[Add rule](#)

Apache is running on <http://34.201.59.201:8084/>

